

Algebra 1
Unit 7
Lesson 5 Practice

Name **Answer Key** _____
Date _____
Period _____

For questions 1-3, factor each quadratic expression using the method of your choice.

1. $10b^2 - 9b - 9$

$$\begin{aligned} &10b^2 - 15b + 6b - 9 \\ &5b(2b - 3) + 3(2b - 3) \\ &(5b + 3)(2b - 3) \end{aligned}$$

2. $12x^2 + 19x - 21$

$$\begin{aligned} &12x^2 + 28x - 9x - 21 \\ &4x(3x + 7) - 3(3x + 7) \\ &(4x - 3)(3x + 7) \end{aligned}$$

3. $6d^2 - 13d + 6$

$$\begin{aligned} &6d^2 - 9d - 4d + 6 \\ &3d(2d - 3) - 2(2d - 3) \\ &(3d - 2)(2d - 3) \end{aligned}$$

For questions 4-5, consider the trinomial $10x^2 - 61x + 72$.

4. Write an expanded form of the trinomial that can be used to factor the expression by grouping.

$$10x^2 - 45x - 16x + 72$$

5. Use the expanded form to factor the expression by grouping.

$$\begin{aligned} &5x(2x - 9) - 8(2x - 9) \\ &(5x - 8)(2x - 9) \end{aligned}$$

For questions 6-7, consider the trinomial $8x^2 + 46x + 45$.

6. Complete the area model.

	$4x$	<u>5</u>
$2x$	$8x^2$	<u>$10x$</u>
<u>9</u>	<u>$36x$</u>	45

7. What are the two binomial factors for $8x^2 + 46x + 45$?

$$(4x + 5)(2x + 9)$$

For questions 8-9, consider the trinomial $10t^2 + 9t - 9$.

8. Complete the area model.

	<u>$5t$</u>	<u>-3</u>
<u>$2t$</u>	<u>$10t^2$</u>	<u>$-6t$</u>
<u>3</u>	<u>$15t$</u>	<u>-9</u>

9. What are the two binomial factors for $10t^2 + 9t - 9$?

$$(5t - 3)(2t + 3)$$

10. Factor $6x^2 + 57x + 132$ using factoring by grouping.

$$3(2x^2 + 19x + 44)$$

$$3(2x^2 + 11x + 8x + 44)$$

$$3(x(2x + 11) + 4(2x + 11))$$

$$3(x + 4)(2x + 11)$$